

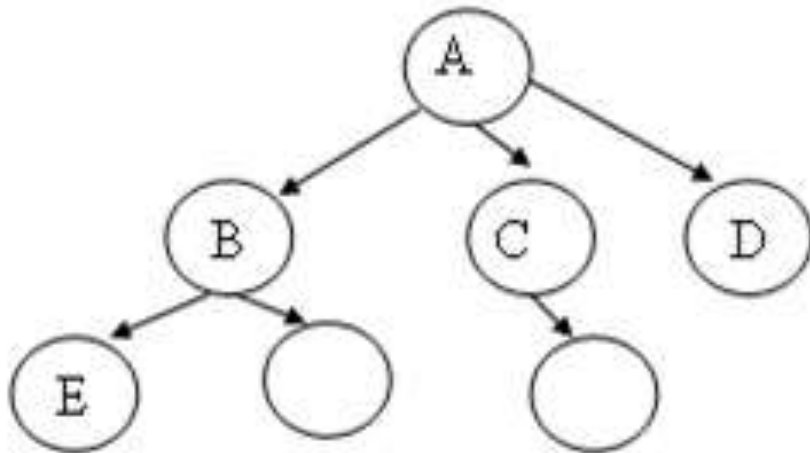
Tree

Tree

- In computer science, a tree is an abstract model of a hierarchical structure.
- It is used to represent a hierarchical relationship existing among several data items.
- A tree consists of nodes with a **parent-child relationship**.
- It is a **non-linear** data structure.
- Searching as fast as **in ordered array**.
- Insertion and deletion **as fast as in linked list**

- Tree is a sequence of nodes.
- There is a starting node known as root node.
- Every node other than the **root** has a parent node.
- Nodes may have any number of children.

Tree



Graph theoretic definition of a Tree

- It is a finite set of one or more data items (nodes) such that
 - There is a special data item called the root of the tree
 - Its remaining data items are partitioned into number of **disjoint subsets**, each of which is itself a tree again, called **subtrees**.

Recursive definition of tree

- A tree t of order n is either empty or consists of a distinguished node r , called the root of T , together with at most n trees, $T_1, T_2, \dots, T_n$ called the sub trees of T .

Some key terms

Degree of a node:

- The degree of a node is the number of children of that node. In above tree the degree of node A is 3.

Degree of a Tree:

- The degree of a tree is the maximum degree of nodes in a given tree.
- In the above tree the node A has maximum degree, thus the degree of the tree is 3.

Path:

- It is the sequence of consecutive edges from source node to destination node. There is a single unique path from the root to any node.

Height of a node:

- The height of a node is the maximum path length from that node to a leaf node. A leaf node has a height of 0.

Height of a tree:

- The height of a tree is the height of the root.

Depth of a node:

- Depth of a node is the path length from the root to that node. The root node has a depth of 0.

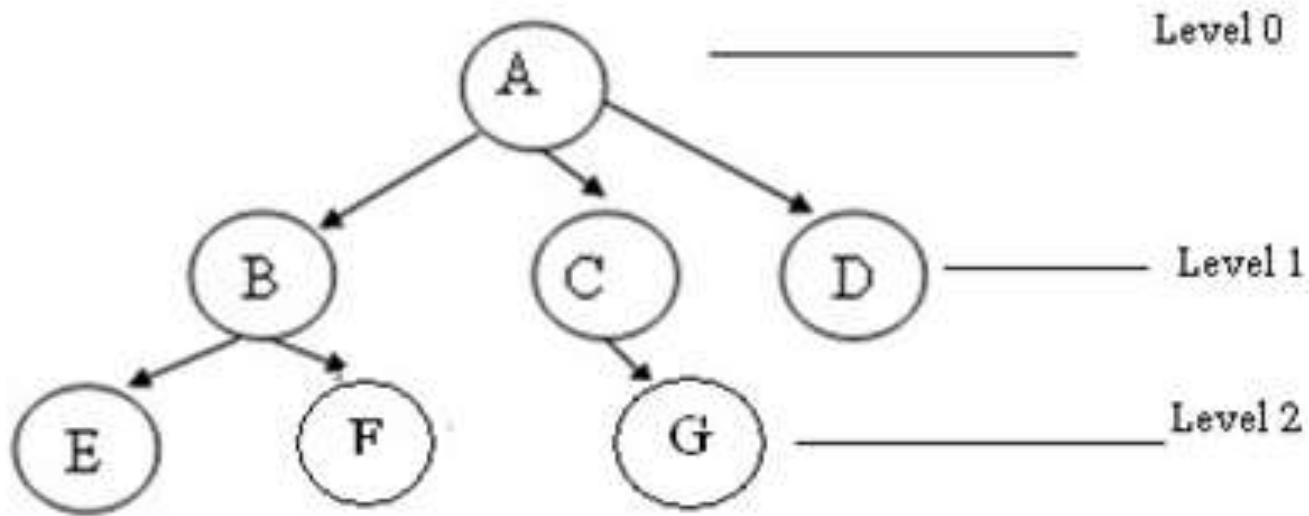
Depth of a tree:

- Depth of a tree is the maximum level of any leaf in the tree. This is equal to the longest path from the root to any leaf.

Level of a node:

- the level of a node is 0, if it is root; otherwise it is one more than its parent.

Illustration:



- A is the root node
- ✓ B is the parent of E and F
- ✓ D is the sibling of B and C
- ✓ E and F are children of B
- ✓ E, F, G, D are external nodes or leaves
- ✓ A, B, C are internal nodes
- ✓ Depth of F is 2
- ✓ the height of tree is 2
- ✓ the degree of node A is 3
- ✓ The degree of tree is 3

Tree traversal

- The tree traversal is a way in which each node in the tree is visited exactly once in a symmetric manner.
- There are three popular methods of traversal
 - ✓ Pre-order traversal
 - ✓ In-order traversal
 - ✓ Post-order traversal

Pre-order traversal

The preorder traversal of a nonempty binary tree is defined as follows:

- ✓ Visit the root node
- ✓ Traverse the left sub-tree in preorder
- ✓ Traverse the right sub-tree in preorder

In-order traversal

- The in order traversal of a nonempty binary tree is defined as follows:
 - ✓ Traverse the left sub-tree in in order
 - ✓ Visit the root node
 - ✓ Traverse the right sub-tree in order

Post-order traversal

- The post-order traversal of a nonempty binary tree is defined as follows:
 - ✓ Traverse the left sub-tree in post-order
 - ✓ Traverse the right sub-tree in post-order
 - ✓ Visit the root node